Predicting Loan Application Cancellations

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1 Introduction

The report analyses loan applications from a Dutch financial institute to predict loan application cancellations (*A_Cancelled*). We chose Application Complete (*A_Complete*) as the decision point to determine whether an application will be cancelled (activity of interest) or reach Offer Accepted (*O_Accepted*). At this decision point, relevant data for prediction is available, and it is still possible for the bank to intervene, making this decision point both appropriate and feasible. Our analysis indicates that a significant proportion of applications move toward cancellation after approximately 27.8 days, likely due to automated timeouts following customer non-response, highlighting the value of early prediction. The proposed model uses credit score, offer count, processing time, loan amount, and activity frequency as key prior features and achieves an accuracy of 84.2%, while also providing insights into the factors influencing cancellation likelihood. The following sections cover our prediction setup and evaluation, followed by a conclusion with our recommendation.

2 Main Part

The prior features used for prediction are credit score, offer count, processing time, loan amount, and activity frequencies, while *A_Complete* serves as the decision point (posterior feature). The prediction task is to determine whether a loan application will be cancelled (treated as a negative outcome) or accepted (treated as a positive outcome). Process discovery shows that after *A_Complete*, applications typically enter the validation phase. Predicting cancellation at this early stage is therefore highly relevant, as it can prevent wasted resources on validation steps for cases that are already on a path to failure. Early prediction also enables the bank to trigger targeted intervention strategies before the customer withdraws, improving both efficiency and customer retention. This prediction task is feasible from a data perspective because all required prior features are consistently recorded before *A_Complete*, and the mutually exclusive outcomes afterwards allow for clean and reliable labelling. It is also feasible from a business perspective, as predictions are made at a point where interventions are still possible, and operational actions can realistically be taken to reduce cancellations and unnecessary validation effort.

To address this task, we implemented a supervised binary classification model using only information available up to application completion. The prior features capture both customer quality and process behavior: credit score strongly reduces cancellation risk (correlation −0.686), processing time increases cancellation likelihood (correlation 0.067), offer count shows weak direct influence (correlation 0.033), and loan amount has a small negative association (correlation −0.023). Activity frequencies were included to reflect behavioral patterns, with some activities showing correlations of up to ≈ 0.14, indicating additional but partial signals of cancellation risk. A feed-forward neural network (Multi-Layer Perceptron) was chosen for prediction because cancellations are driven by non-linear interactions between customer characteristics and process behavior rather than by single factors in isolation. The model is evaluated using a temporal train-test split, where training uses past cases, and testing uses future cases. This evaluation strategy is crucial as it prevents data leakage and ensures that the model does not learn from future information that would not be available in real operational settings. From a business perspective, this setup yields reliable and deployable

predictions that support early identification of high-risk cases, reduce wasted validation effort, and enable timely intervention to prevent avoidable cancellations.

Analysis of the actual process outcomes shows that accepted cases are characterized by stronger and more complete information, as they exhibit a high average credit score of approximately 610, progress through the process relatively quickly with an average processing time of 1.17 days, and involve slightly higher loan amounts (around 18,400). In contrast, cancelled cases are strongly associated with missing or unavailable credit score information (average ≈ 0) and experience longer processing times of about 1.41 days, indicating delays and uncertainty. The number of offers made is broadly similar across both outcomes, suggesting that time delays and information quality, rather than offer frequency alone, are the dominant drivers of cancellation in the real process. The prediction model has successfully learned and internalized these quantitative patterns. It consistently predicts acceptance when credit scores are high (with predicted accepted cases averaging around 774) and processing times are shorter (approximately 1.19 days), while predicting cancellation exclusively in cases with missing credit information and longer processing times (around 1.35 days). This close alignment between learned behavior and actual process dynamics leads to a substantial performance improvement over the naive frequency-based baseline, increasing accuracy from 54.86% to 84.2%, an improvement of 29.40%. From a business perspective, these results demonstrate that cancellations can be anticipated early by monitoring process delays and data completeness, enabling targeted interventions such as accelerating decision steps or resolving missing information, reducing cancellations, improving customer satisfaction, and enhancing overall process efficiency.

3 Conclusion

Predicting cancellations at the *A_Complete* stage is operationally relevant as it represents a key milestone where the bank assesses application viability. The model's primary strength is providing a measurable "lift" over a naive baseline, achieving 84.2% accuracy by capturing process friction such as elapsed time.

However, it must also be noted that the model relies heavily on the CreditScore = 0 feature. Since a zero score represents missing data rather than poor credit, the model is largely predicting based on administrative incompleteness rather than actual customer risk. Furthermore, the model is too conservative; it often flags cases as cancellations only because they take longer than average. This leads to high false-positive rates for customers who are just slow but still intend to sign. Our setup fails to account for the "human element". Since we only used activity counts and basic attributes, we cannot explain why "perfect" customers with high credit scores and fast processing times still cancel their applications. These "False Negatives" are likely driven by factors our model does not see, such as poor phone interactions or better offers from competitors.

To improve outcomes, the business owner should ensure that credit data is fully captured before this milestone to prevent the model from using administrative gaps as a misleading risk indicator. The bank should also investigate the offer generation phase. While not necessarily a trigger, the observed link between multiple offers and cancellations suggests that process complexity at this stage could be a risk factor. Streamlining these iterations toward a "first-time-right" approach could help reduce the churn associated with lengthy negotiations and high process overhead.

4 Prediction details

4.1 Prediction setup

4.1.1 Data Preparation

- **Data Cleaning:** We filtered the event log to include only cases that completed *A_Complete* and reached an outcome (*O_Accepted*, *A_Cancelled*, or *O_Refused*). This ensures the model is trained only on complete process cycles with a definitive decision, avoiding noise from ongoing or incomplete ("ghost") cases and focusing on business-relevant applications.
- **Feature Selection & Engineering:** We used static features (e.g., requested amount, credit score) and engineered dynamic features such as the number of offers created and total time to decision. Together, these capture both customer risk and process friction, helping distinguish whether cancellations stem from applicant characteristics or internal delays.
- **Feature and Prefix Encoding:** We applied state-based bucketing at *A_Complete*, encoding process history using a Bag-of-Words approach and normalizing numerical features with MinMax scaling. This creates a standardized snapshot of each application at a critical decision point, enabling consistent and fair evaluation across all cases.

4.1.2 Modeling

- **Temporal Splitting**: The data was split into training and test sets using a November 1, 2016, cutoff, ensuring the test set contains only cases starting after the training period. We excluded 2,005 cases (6.4%) that crossed this cutoff, as their outcomes occurred later. This setup simulates a real-world deployment and strengthens confidence in the model's ability to predict future loan outcomes.
- **Labeling**: Cases were labeled as "1" (Cancelled/Refused) if the remainder of the trace contained *A_Cancelled* or *O_Refused*, and "0" (Accepted) otherwise. This binary classification directly mirrors the bank's core operational goal: identifying which applications will result in a successful contract. Early detection of cancellation risk allows staff to intervene and offer better terms to high-value clients before they churn.
- **Model Selection**: We implemented a Multi-Layer Perceptron (MLP) with three hidden layers using the ReLU activation function. The MLP architecture can capture the non-linear relationships between credit scores, and process delays that simpler models might miss. The business value lies in higher predictive accuracy, which translates to better-allocated manpower and reduced administrative overhead.

4.1.3 Evaluation Setup

- **Quantitative Metrics**: We evaluated the model using Accuracy, F1-Score, and a Confusion Matrix. While Accuracy provides a general overview of performance, we prioritize the F1-Score to ensure the model effectively identifies the minority class (cancellations). For business, this prevents "metric bias" where high accuracy might hide a failure to detect high-risk customers, ensuring balanced operational decisions.
- **Naive Predictor Comparison**: Our model was benchmarked against a Stratified Naive Predictor to verify that it provides meaningful lift over a simple baseline. This demonstrates that the model captures behavioral patterns rather than merely reflecting historical averages.
- **Qualitative Evaluation:** We analyzed the confusion matrix, focusing on false negatives, to uncover activity patterns behind misclassifications. This improves transparency, helping managers understand model limitations and guiding future process improvements.

4.2 Realization of Prediction Model

The model was realized using a Python-based pipeline leveraging PyTorch for the neural network architecture, pm4py for event log manipulation, and scikit-learn for data preprocessing and baseline comparison. The implementation follows a modular design: a preprocessing script extracts state-based prefixes at the *A_Complete* decision point, followed by an MLP training script utilizing a standard training loop with Adam optimization and Binary Cross-Entropy loss.

4.2.1 Correctness of the prediction and evaluation setup

- Suitable Prediction Model: We utilized a Multi-Layer Perceptron (MLP) because it effectively captures non-linear relationships between static attributes (credit score) and dynamic process features (activity frequency). This allows the bank to detect subtle behavioral triggers for cancellation that simple linear models would overlook, improving risk management.
- No Future Information: Features are engineered strictly from the case history occurring before the *A_Complete* timestamp. This ensures the model remains operationally valid, as it only uses information available to the loan officer at the exact moment a prediction is required.
- Temporal Separation: The dataset was split using a cutoff date (Nov 1, 2016), with earlier cases for training and later cases for testing. This simulates a real-world deployment where the model must predict outcomes for "future" applicants, ensuring the model's reliability against seasonal or policy shifts.
- Correct Labeling: Labels were generated by checking for the presence of *A_Cancelled* or *O_Refused* in the trace suffix after the decision point to define the "Negative Outcome" class, while the presence of *O_Accepted* defined the "Positive Outcome" class. This directly aligns the model with the Key Performance Indicator (KPI) of reducing application churn and identifying likely-to-refuse customers early.
- No Upfront Filtering of Test Cases: The test set includes all cases that reached the decision point, regardless of their complexity or duration. This provides an unbiased view of model performance across the entire customer base, ensuring the bank understands its true error rate.
- Appropriate Evaluation Measure: We used F1-Score and AUC/ROC to account for the class imbalance between accepted and cancelled cases. By prioritizing the F1-score, we balance the cost of "False Alarms" (wasted retention effort) against the risk of "Missed Cancellations" (lost revenue).
- Meaningful Baseline: Our model is compared against a Stratified Naive Predictor that predicts outcomes based on historical class distributions. This proves the "lift" provided by the AI model, justifying the technical investment by showing it performs significantly better than simple guess based on past averages.

4.3 Evidence

4.3.1 Process Model of Decision Point Context

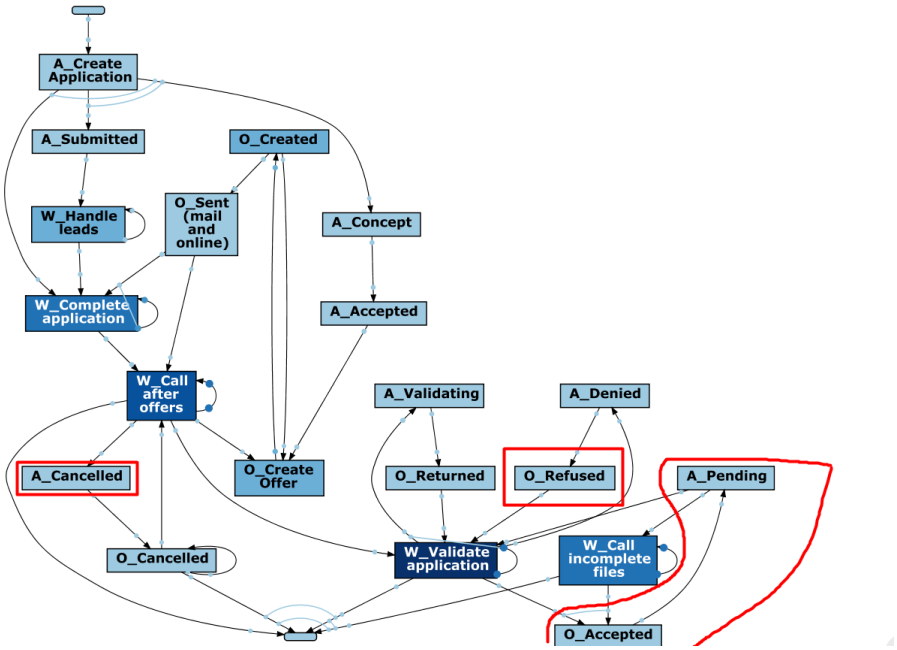

Figure 1: Analysis with iDHM on the whole dataset

Traces	0
Events	0
Event Classes	0
Attributes	0
Variants	0
Events per Trace	0
First Event	292278994-08-17T08:12:55Z
Last Event	292269055-12-02T17:47:04Z

Figure 2: Finding mutually exclusive events by filtering, resulting in zero traces.

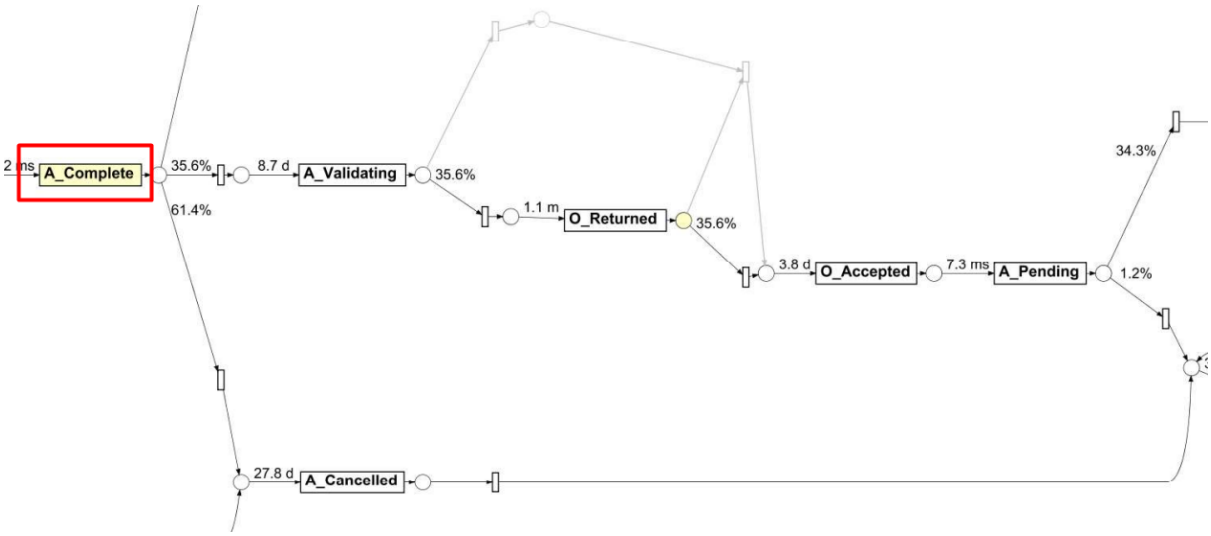

Figure 3: Finding the Decision Point *A_Complete* through Multi-perspective Process Explorer (after generating the Petri Net with iDHM)

Figure 3 shows the process flow surrounding the *A_Complete* activity through Multi-perspective Process Explorer (with average fitness of 99.6%). It illustrates the XOR-split where cases diverge into acceptance or cancellation paths, providing the structural basis for our prediction task. We identified 3 mutually exclusive events (highlighted in Figure 1) by utilizing ProM's Filter Log functionalities (Figure 2): *A_Cancelled* (application cancelled; negative business impact), *O_Refused* (offer refused; negative business impact), *O_Accepted* followed

by *A_Pending* (offer accepted; positive business impact). Filtering the dataset by the presence of either of these events left only 130 traces out of 31,509 unhandled.

Since our goal is to analyze which conditions affect cancellations or successful loan applications and predict which is more likely to occur when, we justified that we can explore 2 model variants: having *O_Accepted* and a) grouping *A_Cancelled* and *O_Refused* under one "negative outcome" label, or b) keeping only *A_Cancelled* as the "negative outcome" and deeming *O_Refused* as an "unclear" outcome and filtering it out with the other 130 traces.

4.3.2 Confusion Matrix

The confusion matrix provides insight into how the prediction model performs across accepted and cancelled applications on the test set. The model performs particularly well in identifying cancelled applications. Out of all cancelled cases, 2,110 are correctly predicted as cancelled, while only 107 cancellations are misclassified as accepted. This corresponds to a high recall of 0.95 for cancellations, indicating that the model rarely misses applications that are likely to fail. From a business perspective, this is desirable, as these cases represent opportunities for early intervention to reduce unnecessary validation effort or prevent customer churn.

As shown in Figure 4, the model correctly classifies 2,025 cases as accepted, while 669 accepted cases are predicted as cancelled. This results in a lower recall of 0.75 for accepted cases but a very high precision of 0.95. This pattern shows that the model adopts a conservative strategy, preferring to flag potentially risky cases even when they eventually result in acceptance. While this leads to additional false positives, it reduces the risk of overlooking true cancellations. The error structure indicates a deliberate trade-off between precision and recall across outcome classes. The model prioritizes capturing cancellation risk over maximizing acceptance recall, which aligns with the business objective of identifying high-risk applications early rather than optimizing for overall classification balance.

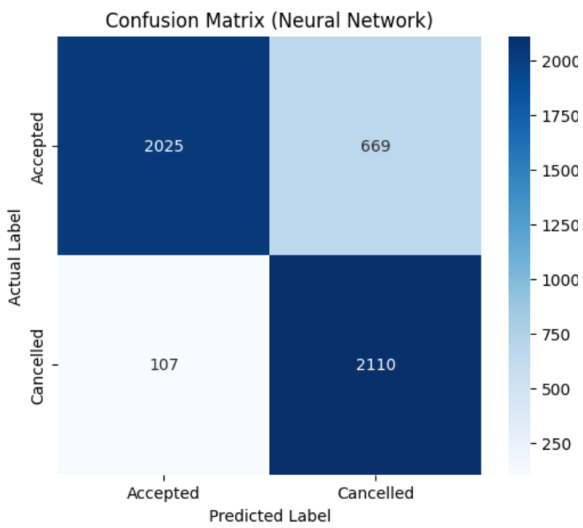

Figure 4: Confusion Matrix

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4.4 Analysis of Prediction Model Quality

4.4.1 Quantitative Analysis

The prediction model achieves an accuracy of 84.2% on the test set, representing an improvement of 29.34% over the naive baseline (as shown in Figure 5). This substantial lift demonstrates that the model captures meaningful relationships between case attributes and

process behaviour rather than relying on historical outcome frequencies. The improvement over the baseline confirms the added value of using a supervised learning approach for early cancellation prediction. In addition, the model achieves an overall F1-score of 0.84 (both macro-averaged and weighted), indicating balanced performance across accepted and cancelled applications despite class imbalance. Overall, these results indicate that the model provides reliable predictive performance in an operational setting, particularly when compared to simple frequency-based prediction.

```
=== RESULTS ===
Model Accuracy: 0.8420
Improvement over Baseline: 29.34%

Classification Report:
              precision    recall  f1-score   support

    Accepted       0.95      0.75      0.84      2694
   Cancelled       0.76      0.95      0.84      2217

    accuracy                           0.84      4911
   macro avg       0.85      0.85      0.84      4911
weighted avg       0.86      0.84      0.84      4911
```

Figure 5: Classification report of the prediction model

4.4.2 Qualitative Analysis

The qualitative evaluation combines attribute-based error analysis and activity-frequency analysis to explain the model's behaviour (as shown in Figure 6). Correctly classified cases are characterized by high credit score availability, moderate offer counts, and average processing times of approximately 1.35 days. In contrast, false positive cases (accepted applications predicted as cancelled) exhibit higher offer counts and slightly shorter processing times, indicating that the model interprets repeated offer generation as a strong cancellation signal. The activity-frequency analysis further supports this interpretation. False positive cases show increased frequencies of offer-related activities such as *O_Created* and *O_Create Offer*, while false negative cases (missed cancellations) are associated with higher frequencies of completion-related activities such as *W_Complete application*. These results show that the model effectively captures process friction but struggles with cancellations that occur despite smooth process execution, highlighting both the strengths and limitations of the current feature set.

```
ACTUAL OUTCOME (0 = Accepted, 1 = Cancelled)
        CreditScore  OfferCount  ProcessingTime        Amount
Actual
0        609.733853    1.124722        1.168338  18427.839644
1          0.000000    1.123139        1.411678  18049.992332

MODEL PREDICTIONS (0 = Predict Accepted, 1 = Predict Cancelled)
           CreditScore  OfferCount  ProcessingTime        Amount
Predicted
0           770.461069    1.071764        1.174473  17933.593809
1             0.000000    1.164088        1.357761  18505.581504

ERROR ANALYSIS
               CreditScore  OfferCount  ProcessingTime        Amount
ErrorType
Correct          397.248609    1.100605        1.305697  17946.905683
False Negative     0.000000    1.046729        1.096314  18882.242991
False Positive     0.000000    1.281016        1.137268  20075.602392
```

Figure 6: Error Analysis.